

Digital Twin and IoT for Smart Campus/Infrastructure

Focus Area: Architecture

R1: Intelligent Campus System Design Based on Digital Twin (2022)

Problem: Fragmented campus systems prevent unified monitoring.

Why this approach: Five-layer architecture: Physical Domain, Virtual Domain, Data Platform, Service Platform, Connection Layer.

Result: Provides real-time synchronization and management.

Review: A layered digital twin architecture integrating IoT, data platforms, and applications for smart campuses.

R2: A Dynamic Digital Twin Framework for Sustainable Facility Management in a Smart Campus (2025)

Problem: Facility management data is distributed across systems.

Why this approach: Combines BIM, GIS, IoT, and Digital Twin technologies.

Result: Improves sustainability monitoring and maintenance.

Review: Demonstrates BIM-GIS-IoT architecture for smart campus operations.

R3: Application of Digital Twin in Campus: Synergistic Effects of IoT, BIM, and GIS

Problem: Campus information is isolated across platforms.

Why this approach: Integrates IoT, BIM, GIS and DT into one architecture.

Result: Creates a unified management platform.

Review: Shows how multiple technologies can be combined into a campus digital twin.

R4: Scalable Digital Twin Deployment in Smart Campus Building Facility Management

Problem: Large-scale deployment is difficult.

Why this approach: Scalable architecture for campus facilities.

Result: Supports expansion across multiple buildings.

Review: Useful reference for deployment architecture.

R5: A Web of Things Architecture for Digital Twin Creation and Model-Based Reinforcement Control

Problem: Interoperability between devices is limited.

Why this approach: Uses Web of Things architecture.

Result: Improves connectivity and interoperability.

Review: Demonstrates standards-based digital twin architecture.

R6: A Comprehensive Review of Digital Twin – Part 1

Problem: Lack of unified understanding of DT architectures.

Why this approach: Reviews common DT frameworks and architectures.

Result: Identifies architecture patterns.

Review: Useful foundation paper for taxonomy development.

R7: Digital Twin Framework for Facility Management of Modular Buildings

Problem: Facility data integration is difficult.

Why this approach: Combines BIM, IoT and DT architecture.

Result: Improves facility management workflows.

Review: Applicable to campus infrastructure.

R8: Architecting Digital Twins for Intelligent Transportation Systems

Problem: Transportation infrastructure requires real-time monitoring.

Why this approach: Modular digital twin architecture.

Result: Supports analytics and monitoring.

Review: Useful infrastructure architecture example.

R9: Design of Intelligent Manufacturing System Based on Digital Twin

Problem: Manufacturing systems require synchronization.

Why this approach: Industrial digital twin architecture.

Result: Supports monitoring and control.

Review: Provides transferable architecture concepts.

R10: Digital Twin and BIM-Based Intelligent Dispatching Platform

Problem: Construction management lacks integrated information.

Why this approach: BIM-centered digital twin architecture.

Result: Improves information integration and decisions.

Review: Strong example of BIM-DT architecture.

Final Review

The reviewed papers show four major architecture categories: layered architectures, BIM-IoT architectures, BIM-GIS-IoT architectures, and Web/Cloud-based architectures. Across the literature, successful smart campus architectures rely on real-time IoT sensing, digital twin synchronization, scalable data platforms, and decision-support applications. The most common trend is the integration of BIM and IoT into a digital twin platform that supports monitoring, facility management, and sustainability objectives.